

PAGE: Progressive Anomaly Generation Network for Semi-supervised Graph Anomaly Detection

Ting Guo
gting@nuc.edu.cn
North University of China
Taiyuan, Shanxi, China

Dongyu Pei
North University of China
Taiyuan, Shanxi, China
sz202407052@st.nuc.edu.cn

Gangzhu Qiao*
Qiaogz@nuc.edu.cn
North University of China
Taiyuan, Shanxi, China

Kaixuan Yao
Shanxi University
Taiyuan, Shanxi, China
ykx@sxu.edu.cn

Da Wang*
wanda@sxu.edu.cn
Shanxi University
Taiyuan, Shanxi, China

Abstract

Semi-supervised graph anomaly detection confronts the fundamental challenge of identifying anomalous nodes that exhibit deviations from normal graph patterns in node attributes or structure connectivity, using a small set of labeled normal nodes. Existing methods fall short in modeling the spectrum of anomaly severity, as they generate anomalies in a single, undifferentiated step. This oversight restricts the detection of complex anomalies. In this paper, we propose a progressive anomaly generation network (PAGE) to overcome this limitation. PAGE enhances complex anomaly detection through progressive attribute-structure perturbation: It first injects attribute noise into selected normal nodes to generate pseudo-anomalous nodes. Then it applies progressive structure perturbation to create hybrid anomalies that simulate the evolution from mild to complex perturbations. Importantly, the proposed anomaly progressive constraint loss enforces that hybrid anomalies exhibit a higher degree of abnormality than primary anomalies, enhancing the model's anomaly quantification capability. PAGE further integrates deviation constraint loss and classification loss to optimize node representations and improve prediction. Extensive experiments on benchmark datasets demonstrate that PAGE significantly outperforms state-of-the-art methods, demonstrating its effectiveness for graph anomaly detection.

CCS Concepts

• Computing methodologies → Machine learning.

Keywords

graph neural network, semi-supervised graph anomaly detection, graph anomaly detection

ACM Reference Format:

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*Corresponding author (Equal contribution).



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1 Introduction

Graph anomaly detection (GAD) has attracted growing research interest due to its broad applications in network security [11], financial risk control [19], and social analysis [32]. This technique identifies nodes exhibiting abnormal structural relations [15] or attribute features [6]. Its core mechanism detects local inconsistencies between nodes and their neighborhoods in either feature distributions or connection patterns. An anomalous node is identified when its attributes or interactions significantly deviate from expected behavior patterns within its neighborhood structure [29, 37].

Current research on GAD primarily focuses on two paradigms: unsupervised and semi-supervised. Unsupervised GAD assumes that the ground-truth labels of all nodes in the training data are unknown [1, 8]. Its core idea is to learn the inherent patterns or representations of graph data (including node attributes and topological structure), thereby identifying anomalous nodes that deviate from these patterns. Representative technical approaches include reconstruction-based methods [8, 17] and contrastive learning-based methods [2, 10, 34, 42]. Semi-supervised GAD assumes the availability of known label information for a subset of nodes in the training data [3, 14, 18]. Depending on the composition of known labels, semi-supervised GAD can be further categorized into two types: mixed label type and normal-only label type. Mixed label type assumes that the known labeled samples include both normal and anomalous nodes [14, 18, 25, 26, 33]. The normal-only label type assumes that the known labeled samples consist solely of normal nodes [30]. Although the mixed-label paradigm provides more comprehensive supervisory information, acquiring reliable annotations for anomalous nodes is often extremely difficult and costly in practical application scenarios. In contrast, identifying and annotating substantial normal samples is typically more feasible. Hence, the semi-supervised paradigm utilizing only normal node labels offers stronger practical applicability. This paper focuses specifically on and conducts an in-depth investigation into the semi-supervised GAD problem under the “normal-only labels” setting.

In semi-supervised GAD, generative methods have gained traction by generating pseudo-anomalous nodes to mitigate label scarcity. GGAD [30] synthesizes anomalies through attribute perturbation and refines representations solely via neighbor affinity. CGenGA

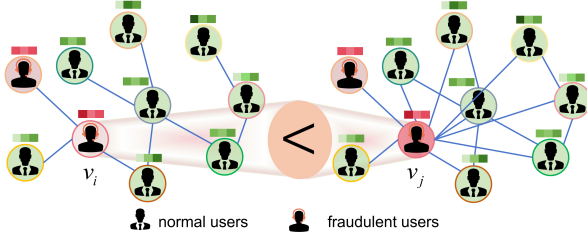


Figure 1: A diagram of motivation. The anomaly degree of node v_i is less than that of node v_j .

[26] and LEDGM [40] focus on reconstructing original data distributions—the former via denoising principles and the latter through joint latent representation optimization. While Dagad [24], CONSIGAD [3] and gADAM [43] perform data augmentation via embedding-space interpolation. Although some existing unsupervised and semi-supervised approaches have explored the synergy between node attributes and graph structure, they still fall short in modeling the spectrum of anomaly severity. These methods typically generate anomalous nodes in a single, undifferentiated step, failing to capture the continuous variation in deviation degrees observed in real anomalies. This limitation severely hinders their ability to effectively detect complex anomalies. As shown in Figure 1, node v_i only exhibits attribute anomalies, while node v_j demonstrates both attribute anomalies and significant structural deviations (its node degree is much higher than normal nodes). This aligns with real-world scenarios: in social networks, normal users typically connect with dozens of friends, whereas fraudulent users (such as bot or marketing accounts) may suddenly establish thousands of connections, displaying dual anomalies in both attributes and structure. However, most existing methods struggle to model such continuous variation in anomaly degrees, failing to effectively quantify and perceive differences in anomaly intensity among nodes.

To address this issue, we conceptualize anomaly generation as a two-stage, progressive process. By constructing a hierarchical set of pseudo-anomalies, our approach can more accurately approximate the full anomaly spectrum and explicitly quantify different levels of anomalous deviation. This fine-grained anomaly generation strategy significantly enhances the model’s capability to identify complex anomalous patterns. Therefor, in this paper, we propose a progressive anomaly generation network (PAGE) for semi-supervised GAD. PAGE integrates progressive attribute-structure perturbation with an anomaly progressive constraint loss. This approach aims to enhance the model’s ability to capture complex anomalous behaviors and quantify anomaly severity. The method first injects attribute noise into a subset of normal nodes to generate primary anomalous embeddings. It then applies structural perturbations to produce hybrid anomalous embeddings. This process simulates the progressive evolution of anomalous behaviors observed in real-world scenarios. To enhance the model’s ability to quantify the degree of anomalies, we designed an anomaly progressive constraint loss. This loss forces attribute perturbations to occur before structural perturbations and quantifies the hierarchical degradation process by utilizing distance constraints. This allows the model to both simulate complex anomalies more realistically

and effectively quantify different degrees of anomalies. Accordingly, our main contributions can be summarized as follows:

- We propose a novel semi-supervised GAD framework, PAGE, which designs progressive attribute-structure perturbations to capture complex anomalous behaviors.
- An anomaly progressive constraint loss is designed to enforce the condition that hybrid anomalies exhibit a higher degree of abnormality than primary anomalies, thereby enhancing the model’s anomaly quantification capability.
- Extensive experiments demonstrate that PAGE achieves superior detection performance on benchmark datasets.

2 Related Work

2.1 Unsupervised Graph Anomaly Detection

Unsupervised GAD assumes that the ground-truth labels of all nodes in the training data are unknown [1, 8]. Representative technical approaches include reconstruction-based methods [8, 17, 31] and contrastive learning-based methods [2, 10, 34, 42]. We further argue that these approaches focus on two key aspects: attributes and structure. Attribute-oriented methods identify anomalies by detecting deviations in node features, often employing techniques like clustering [41], density estimation [36], or autoencoder [22] to pinpoint outliers within the attribute space [2, 7, 8, 10, 31, 34, 42]. However, purely attribute-driven methods often ignore the structural context information encoded in node neighborhoods, which limits their effectiveness in scenarios where structural dependencies are crucial. Conversely, structure-oriented methods [8, 20, 28] focus on models in the graph topology, utilizing structural similarity, connectivity pattern analysis, or subgraph modeling to uncover anomalies manifested in the graph’s connectivity.

2.2 Semi-supervised Graph Anomaly Detection

Semi-supervised GAD leverages both labeled and unlabeled nodes, with most approaches utilizing labeled samples from both normal and anomalous classes. Existing semi-supervised GAD methods fall into two primary categories: **Attribute-oriented methods** focus on exploiting node feature patterns. Among these, CGenGA [26] establishes two fundamental principles for designing denoising neural networks specifically tailored to graph anomaly generation. LEDGM [40] introduces a novel end-to-end deep generative model that conditions anomaly label prediction directly on learned latent representations, explicitly optimizing for detection performance rather than data reconstruction fidelity. DagAD [24] framework employs three specialized modules (structure extractor, attribute augments, and anomaly detector) for comprehensive data augmentation in attributed graphs. gADAM [43] enhances training data through adaptive interpolation of data instances in the embedding space. **Structure-oriented methods** concentrate on topological modeling. AO-GNN [18] develops an AUC-oriented reinforcement learning module guided by surrogate AUC rewards to search for optimal edge pruning policies. BHetero-GHRN [14] addresses graph heterogeneity by pruning inter-class edges through strategic emphasis and delineation of high-frequency graph components. PC-GNN [25] implements a neighborhood sampler that selectively oversamples minority-class neighbors while under-sampling majority-class neighborhoods to address class imbalance. H²-FDetector [33]

explicitly models both homophilic and heterophilic interactions through dedicated network pathways in its GNN architecture. Notably, existing methods typically assume access to labeled samples from both normal and anomalous classes, exhibiting limitations under the setting where only labeled normal samples are provided. For instance, GGAD [30] synthesizes pseudo-anomalies through attribute perturbation and refines node representations via neighborhood affinity optimization. This paper specifically targets the challenging scenario of having only labeled normal samples. Current semi-supervised GAD approaches either employ isolated attribute perturbation or leverage structure-agnostic embedding synthesis, failing to explicitly model the synergistic relationship between attributes and structure during anomaly generation. Our PAGE enhances the detection of complex anomalies through progressive joint attribute-structure perturbation, thereby achieving a synergistic enhancement effect where both attributes and structure collaboratively contribute to anomaly detection.

3 Method

3.1 Problem Definition

Let $G = (V, E, \mathbf{X})$ be a graph, where $V = \{v_1, v_2, \dots, v_n\}$ denotes the set of nodes, $E \subseteq V \times V$ is the set of edges, and $\mathbf{X} \in \mathbb{R}^{n \times m}$ is the node feature matrix, where the i -th row $x_i \in \mathbb{R}^m$ represents the attribute vector of node v_i . The adjacency matrix is denoted as $\mathbf{A} \in \{0, 1\}^{n \times n}$, where $A_{ij} = 1$ indicates that there exists an edge $(v_i, v_j) \in E$, and $A_{ij} = 0$ otherwise. GAD aims to identify nodes that significantly deviate from the majority in terms of structural and/or attribute patterns. In the semi-supervised setting, we have access to a limited set of labeled normal nodes $V_l \subset V$.

Node embeddings are generated through a two-layer graph convolutional network (GCN) [5, 13, 16]. The model takes the node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and the adjacency matrix $\mathbf{A} \in \{0, 1\}^{n \times n}$ (including self-loops) as inputs. These are then passed through two GCN layers. Each layer captures one-hop neighborhood information, expands the receptive field, and enables the fusion of local structure and attribute information. Each GCN layer is computed as:

$$\mathbf{H}^{(l)} = \text{GNN}(\mathbf{H}^{(l-1)}, \mathbf{A}) = \phi\left(\mathbf{D}^{-\frac{1}{2}} \mathbf{A} \mathbf{D}^{-\frac{1}{2}} \mathbf{H}^{(l-1)} \Theta^{(l)}\right), \quad (1)$$

where $\mathbf{D}$ is the degree matrix of $\mathbf{A}$, $\Theta^{(l)}$ is the weight matrix, and $\phi(\cdot)$ denotes a non-linear activation function, l is the layer hyper-parameter. We define $\mathbf{H}^{(0)} = \mathbf{X}$ as the input, the output is $\mathbf{H}$.

This paper focuses on semi-supervised GAD with only normal node labels. We construct pseudo-anomalous nodes by injecting noise into partially labeled nodes, then integrate these pseudo-anomalous nodes with normal nodes to train a binary classifier. The key challenges are (1) How to select appropriate noise types. (2) How to enhance the model's capability to discern complex anomaly patterns. (3) How to design losses to improve the quality of anomalous nodes and enhance the model's ability to quantify anomalies.

3.2 PAGE Framework Overview

To address the challenges above, we propose the PAGE, as shown in Figure 2. For **challenge 1**, the model introduces both attribute noise and structural noise. Specifically, attribute noise is used to simulate outliers in the feature space (e.g., anomalous attribute

values), while structural noise is employed to model topological anomalies (e.g., cross-community links). For **challenge 2**, PAGE implements a hierarchical noise injection strategy on a set of normal node embeddings. In the primary phase, attribute noise is exclusively introduced through Gaussian perturbations applied to node features, generating primary anomalous embeddings that simulate isolated feature-space deviations. Subsequently, structural noise is strategically superimposed via stochastic edge rewiring operations, synthesizing hybrid anomalous embeddings that emulate complex composite anomalies. This proactive strategy for generating anomalous embeddings enables the model to learn to identify diverse anomaly patterns ranging from simple to complex. For **challenge 3**, PAGE introduces a deviation constraint loss and an anomaly progressive constraint loss. The deviation constraint loss is designed to keep the generated primary anomaly nodes within a plausible range while preserving adequate separability. The anomaly progressive constraint loss enforces a progressive distance relationship among the generated primary anomalous embeddings, hybrid anomalous embeddings, and normal embeddings (namely, hybrid anomalous embeddings should lie farther from normal embeddings than primary anomalous embeddings). These two losses work synergistically to significantly enhance the model's ability to identify real-world anomaly patterns.

3.3 Noise Type Selection

PAGE framework strategically integrates attribute perturbation and structural perturbation. Firstly, after getting $\mathbf{H}$ through GNN_1 , we select a subset of nodes $\mathcal{S} \subset \mathcal{V}$, $|\mathcal{S}| = k$ and their induced subgraph $\mathbf{A}_{\mathcal{S}}$ from the labeled normal nodes. We fixedly select a few normal labeled nodes.

For attribute perturbation, we simulate feature-space outliers through Gaussian perturbations:

$$\hat{\mathbf{S}} = \mathbf{S} + \epsilon, \quad \epsilon = \sigma \cdot \mathcal{N}(0, \mathbf{I}) + \mu. \quad (2)$$

For structural perturbation, we model topological anomalies via edge rewiring:

$$\mathbf{A}_{\hat{\mathcal{S}}} = \mathbf{A}_{\mathcal{S}} \odot \mathbf{M}_{\mathcal{S}} + (1 - \mathbf{A}_{\mathcal{S}}) \odot \mathbf{M}_{\mathcal{P}}, \quad (3)$$

where $\mathbf{M}_{\mathcal{S}}, \mathbf{M}_{\mathcal{P}} \sim \text{Bernoulli}(p_{\mathcal{S}})$. This operation deletes existing edges with $\mathbf{M}_{\mathcal{S}}$ (mask probability $p_{\mathcal{S}}$) and adds cross-community edges with $\mathbf{M}_{\mathcal{P}}$ creating pathological connections between topologically distant nodes (e.g., bridging social communities).

3.4 Progressive Anomaly Embedding Generation

Anomalies in real-world datasets are often triggered by a combination of multiple factors (e.g., both anomalous user attributes and anomalous relationships in social networks, as well as both anomalous amounts and anomalous fund flows in financial transactions). We utilize a staged operation by first injecting attribute noise and then superimposing structural noise precisely to model the real formation process of such multi-factor anomalies. Given attribute-perturbed embeddings $\hat{\mathbf{S}}$ and subgraph adjacency $\mathbf{A}_{\mathcal{S}}$:

$$\hat{\mathbf{S}}' = \text{GNN}_2(\hat{\mathbf{S}}, \mathbf{A}_{\mathcal{S}}), \quad (4)$$

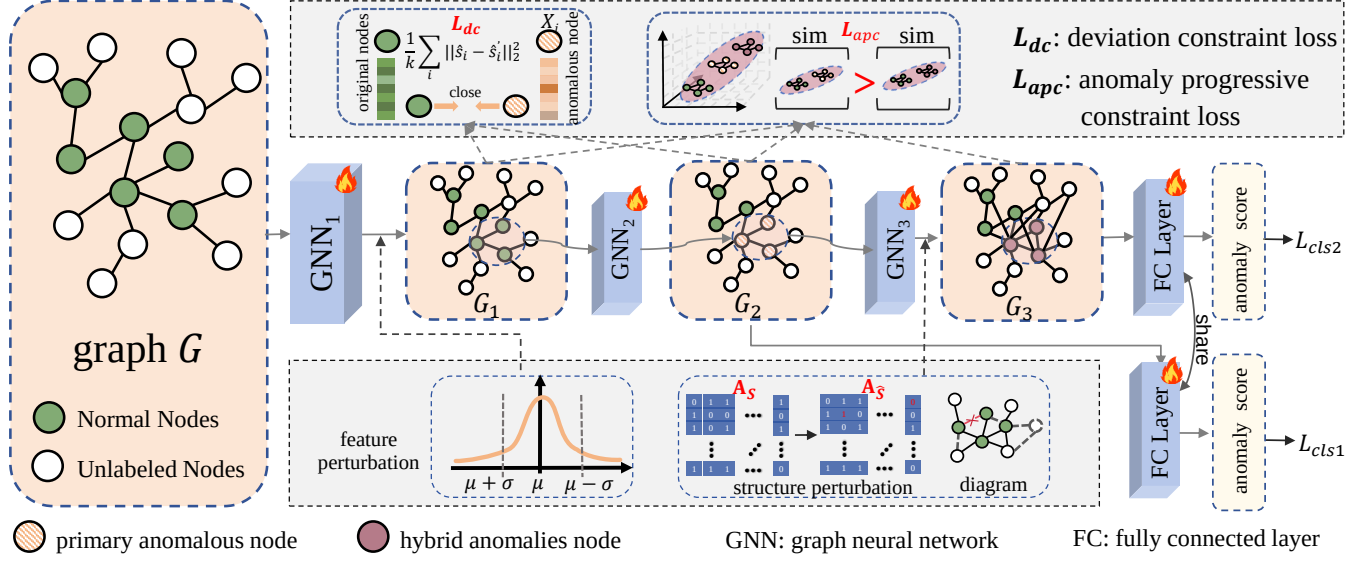


Figure 2: The overall framework of PAGE. PAGE enhances complex anomaly detection by progressive attribute-structure perturbation: it first injects attribute noise into selected normal nodes to generate pseudo-anomalous nodes. It then applies progressive structural perturbations to create hybrid anomalies, modeling the evolution from minor to complex perturbations. Finally, PAGE integrates the generated pseudo-anomaly nodes, normal nodes, and unlabeled nodes to train the classifier, see FC Layer in the figure.

where $\hat{S}'$ is **primary anomalous embeddings**. Propagate primary anomalous embeddings through perturbed topology:

$$\hat{S}'' = GNN_3(\hat{S}', A_{\hat{S}}), \quad (5)$$

where $\hat{S}''$ is **hybrid anomalous embeddings**. This design enables the model to not only identify simple anomalies caused by a single factor, but also effectively detect complex anomalous nodes where attribute and structural anomalies are intertwined, significantly improving the model's generalization ability in real scenarios.

3.5 Optimizing Pseudo-Anomaly Quality via Loss Regularization

The **deviation constraint loss** is designed to ensure that the generated primary anomaly nodes maintain a “plausible and separable deviation” from their normal counterparts. In GAD, real anomalous nodes often exhibit subtle attribute deviations while still remaining close to the normal feature distribution, a phenomenon referred to as “feature camouflage”. However, existing methods struggle to accurately model such fluctuating deviations in a controlled manner. To address this, we introduce a deviation constraint loss that not only mimics realistic attribute noise but also preserves the discriminability between generated anomalies and normal samples. This is achieved by constraining the distance between the original node embeddings $\hat{s}_i$ and their perturbed ones $\hat{s}'_i$, thereby ensuring that pseudo-anomalies do not diverge excessively while still retaining meaningful feature-level diversity. The loss is formulated as:

$$\mathcal{L}_{dc} = \frac{1}{|k|} \sum_i \|\hat{s}_i - \hat{s}'_i\|_2^2, \quad (6)$$

where $\hat{s}_i \in \hat{S}$ denotes the original node embedding, $\hat{s}'_i \in \hat{S}'$ represents the anomalous ones, and $|k|$ is the number of selected normal nodes. This formulation allows for controlled deviation, encouraging the model to generate subtle yet discernible anomalies. By preventing the generated embeddings from being too distant from the normal region, the loss avoids generating unrealistic outliers, while the introduced noise variations enhance sample diversity.

In addition, we design the **anomaly progressive constraint loss**, which enforces a progressive distance relationship between the primary anomaly embedding, the hybrid anomaly embedding, and the normal embedding. Primary anomalous embedding represents anomalous nodes with only attribute perturbation, while hybrid anomaly embedding encompasses both attribute perturbation and structural perturbation, resulting in more complex anomalous semantics. To significantly enhance the model's ability to quantify anomaly severity, we design an anomaly progressive constraint loss. The core principle of this loss is that semantically simpler anomalies (e.g., those with only feature perturbation) should reside closer to the region of normal nodes, reflecting their mild deviation from the normal state. Conversely, semantically complex hybrid anomalies (e.g., hybrid anomalies) should maintain a greater distance from normal nodes to capture their more severe deviations. This progressive distance constraint enables the model to distinguish the severity levels of different anomalies with precision. The anomaly progressive constraint loss is defined as:

$$\mathcal{L}_{apc} = \frac{1}{|k|} \sum_i \max(0, m + \|s_i - \hat{s}'_i\|_2 - \|s_i - \hat{s}''_i\|_2), \quad (7)$$

where $s_i \in S$, $\hat{s}'_i \in \hat{S}'$, $\hat{s}''_i \in \hat{S}''$, m is the hyperparameter. Enhancing the model's ability to quantify anomaly severity is crucial. This

capability helps reveal the impact of different perturbation types and their combinations on anomaly severity, thereby increasing the model’s sensitivity to subtle shifts in anomaly distributions. Thus, the introduction of the margin loss equips the model with the key ability to comparatively evaluate anomaly severity, ultimately enhancing its anomaly detection capability.

3.6 Overall Loss

To better discriminate between normal and anomalous nodes, this paper designs a dual classification loss. After obtaining pseudo-anomalous nodes, we execute a classifier training procedure. Specifically, we train the same classifier in two sequential phases: First, leveraging primary anomalous embeddings along with embeddings of normal nodes and unlabeled nodes to train a classifier. The classification loss is $\mathcal{L}_{\text{cls}_1}$. Subsequently, training the identical classifier using hybrid anomaly embeddings, similarly integrated with normal and unlabeled node embeddings. The classification loss is $\mathcal{L}_{\text{cls}_2}$. This design ensures that the classifier progressively learns to distinguish anomalies of distinct complexities. Consequently, the model achieves robust classification performance against anomalous signals from diverse sources and complexity levels. Thus, the final optimization objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}_1} + \lambda_1 \mathcal{L}_{\text{cls}_2} + \lambda_2 \mathcal{L}_{\text{dc}} + \lambda_3 \mathcal{L}_{\text{apc}}, \quad (8)$$

where λ_1 , λ_2 and λ_3 are hyperparameters that control the relative importance of each objective during training.

4 Experiments

4.1 Experimental Setup

Datasets. To better simulate real-world anomaly detection scenarios, we evaluate our method on six public datasets spanning diverse domains, including social networks, e-commerce co-purchase graphs, blockchain transactions, and financial networks. Specifically, Reddit [21] is a social interaction graph; Photo [27] and Amazon [9] are product co-purchase networks; Elliptic [39] records Bitcoin transactions; T-Finance [35] is a financial knowledge graph [23]; and DGraph [19] comes from a large-scale financial network. During training, we adopt a semi-supervised strategy using only a subset of labeled normal nodes for supervision, while most nodes remain unlabeled, better reflecting real-world scenarios where labeled anomalies are typically scarce and incomplete.

Compared Methods and Evaluation Metrics. To evaluate the effectiveness of the PAGE, we conduct comparative experiments under the semi-supervised setting described above. Seven representative GAD algorithms are selected as baselines. DOMINANT [8] and AnomalyDAE [12] are unsupervised graph autoencoder methods. OCGNN [38] and TAM [28] adopt one-class learning to identify anomalies via compact normal subspaces or affinity constraints. AEGIS [7] and GAAN [4] leverage graph generation frameworks, with AEGIS combining contrastive and reconstruction signals, and GAAN incorporating adversarial mechanisms. GGAD is a recent semi-supervised generative method that synthesizes anomalies through attribute perturbation and refines representations using neighbor affinity. We evaluate all methods using AUROC, which measures the overall discriminative ability across thresholds, and

AUPRC, which focuses on the detection performance of anomalies, with higher values indicating better performance.

Implementation Details. All experiments are conducted on a server equipped with an NVIDIA RTX 4090 GPU, using PyTorch 1.11.0 with CUDA 11.5. The model is trained using the Adam optimizer. To simulate the scarcity of anomaly samples in real-world scenarios, a subset of k normal nodes is randomly selected for supervision, while the remaining nodes remain unlabeled, thereby forming a semi-supervised learning setting. The selected node ratios correspond to $\{0.2, 0.3, 0.5, 0.6, 0.8\}$, and the results for ratio 0.5 are reported in Table 1. To improve the model’s robustness against uncertainty in real-world data, Gaussian noise (σ, μ) is injected into node features, and structural perturbations are applied by randomly adding and deleting edges in the adjacency matrix at a certain proportion. In addition, a hyperparameter m (margin) is introduced to constrain that the embedding distance between hybrid anomalies and normal nodes is at least m greater than that between primary anomalies and normal nodes, ensuring sufficient separation among different anomaly levels in the embedding space. All experiments are repeated five times with random seeds, and the averaged results are reported to demonstrate stability and reproducibility. The source code of PAGE has been available on GitHub¹.

4.2 Overall Performance Analysis

Table 1 provides a comprehensive comparison of various GAD methods under both unsupervised and semi-supervised settings across six real-world datasets. Overall, our method PAGE achieves state-of-the-art performance, significantly outperforming existing baselines on most datasets – particularly in scenarios involving structural noise and complex anomaly patterns. Notably, PAGE performs exceptionally well on datasets where anomalies are closely related to topological irregularities.

On the Photo dataset, which involves detecting fake product associations, PAGE achieves an AUROC of 0.8094 and an outstanding AUPRC of 0.5874, far surpassing all competitors. On T-Finance, a graph characterized by inherent structural noise, PAGE attains an AUPRC of 0.6521, representing a 257% improvement over the strongest baseline, GGAD (0.1825). On the Elliptic temporal transaction graph, which contains illicit financial activities, PAGE improves AUROC by 10% (0.8019 vs. GGAD’s 0.7290), demonstrating better robustness in dynamic and noisy environments. These results indicate that PAGE effectively handles structurally perturbed anomaly patterns, particularly when the injected noise aligns with real-world anomaly distributions. By synergistically integrating feature reconstruction with topological perturbation, PAGE enhances the discrimination between normal and anomalous nodes under the semi-supervised GAD setting.

In cases where anomalies are more attribute-oriented, PAGE remains highly competitive. On the Amazon graph, GGAD slightly outperforms PAGE in AUROC (0.9443 vs. 0.9425), due to its design compatibility with stable topological associations. However, PAGE surpasses GGAD in AUPRC (0.8098 vs. 0.7922), highlighting the strength of its anomaly progressive constraint loss in detecting subtle feature-based anomalies, such as fraudulent high-frequency reviews. On Reddit social graph, PAGE achieves a record AUROC

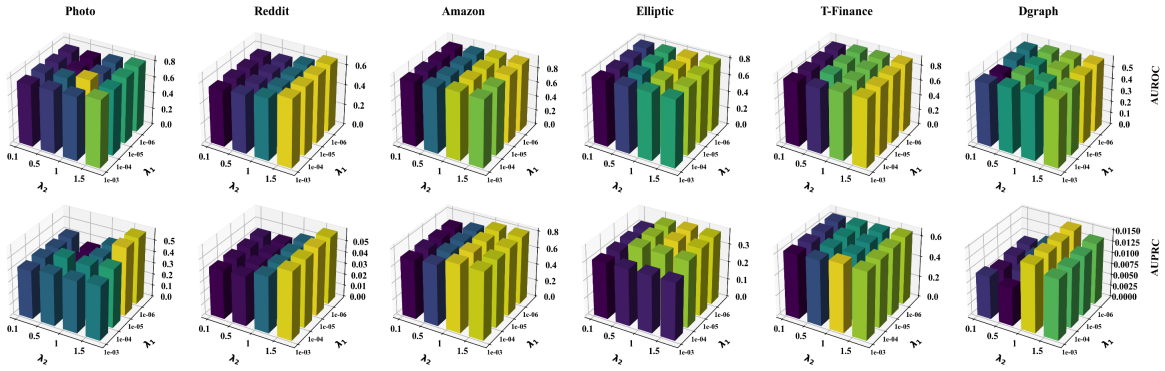
¹<https://github.com/TingGuo301/PAGE>

Table 1: Performance comparison of different GAD methods on six datasets. We bold the highest scores.

Setting	Method	Dataset											
		AUROC						AUPRC					
		Photo	Reddit	Amazon	T-Finance	Elliptic	DGraph	Photo	Reddit	Amazon	T-Finance	Elliptic	DGraph
Unsupervised	Dominant	0.5136	0.5105	0.7025	0.6087	0.2960	0.5738	0.1039	0.0380	0.1315	0.0536	0.0454	0.0075
	AnomalyDAE	0.5069	0.5091	0.7783	0.5809	0.4963	0.5763	0.0987	0.0319	0.1429	0.0491	0.0872	0.0070
	OCGNN	0.5307	0.5246	0.7165	0.4732	0.2581	/	0.0965	0.0375	0.1352	0.0392	0.0616	/
	AEGIS	0.5516	0.5349	0.6059	0.6496	0.4553	0.4509	0.0972	0.0413	0.1200	0.0622	0.0827	0.0053
	GAAN	0.4296	0.5216	0.6513	0.3091	0.2590	/	0.0767	0.0348	0.0852	0.0283	0.0436	/
	TAM	0.5675	0.6062	0.8303	0.6175	0.4039	/	0.1013	0.0370	0.4024	0.0547	0.0502	/
Semi-supervised	Dominant	0.5314	0.5194	0.8867	0.6167	0.3256	0.5851	0.1238	0.0414	0.7289	0.0542	0.0652	0.0076
	AnomalyDAE	0.5272	0.5280	0.9171	0.6027	0.5409	0.5866	0.1177	0.0362	0.7748	0.0538	0.0949	0.0071
	OCGNN	0.6461	0.5622	0.8810	0.5742	0.2881	/	0.1501	0.0400	0.7538	0.0492	0.0640	/
	AEGIS	0.5936	0.5605	0.7593	0.6728	0.5132	0.4450	0.1110	0.0441	0.2616	0.0685	0.0912	0.0058
	GAAN	0.4355	0.5349	0.6531	0.3636	0.2724	/	0.0768	0.0362	0.0856	0.0324	0.0611	/
	TAM	0.6013	0.5829	0.8405	0.5923	0.4150	/	0.1087	0.0446	0.5183	0.0551	0.0552	/
	GGAD	0.6476	0.6354	0.9443	0.8228	0.7290	0.5943	0.1442	0.0610	0.7922	0.1825	0.2425	0.0082
	PAGE (Ours)	0.8094	0.6710	0.9425	0.9001	0.8019	0.5954	0.5874	0.0615	0.8098	0.6521	0.3748	0.0132

Table 2: Evaluation of various loss function configurations in semi-supervised GAD on six benchmark datasets. The best results for each metric are highlighted in bold.

Method	Dataset											
	AUROC						AUPRC					
	Photo	Reddit	Amazon	Elliptic	T-Finance	DGraph	Photo	Reddit	Amazon	Elliptic	T-Finance	DGraph
$\mathcal{L}_{cls_1}$	0.6149	0.5224	0.7489	0.7890	0.8567	0.5308	0.1207	0.0398	0.2170	0.3530	0.2875	0.0061
$\mathcal{L}_{cls_1} + \mathcal{L}_{dc}$	0.7030	0.6377	0.8118	0.8001	0.4912	0.5505	0.1807	0.0560	0.4778	0.3565	0.0409	0.0079
$\mathcal{L}_{cls_1} + \mathcal{L}_{dc} + \mathcal{L}_{apc}$	0.8167	0.6673	0.9331	0.8010	0.8762	0.5516	0.5571	0.0591	0.8070	0.3782	0.6119	0.0082
$\mathcal{L}_{cls_2}$	0.6113	0.5461	0.7668	0.6745	0.8363	0.5417	0.1192	0.0412	0.2542	0.1356	0.1898	0.0081
$\mathcal{L}_{cls_2} + \mathcal{L}_{dc}$	0.6871	0.6510	0.8404	0.8010	0.4893	0.5482	0.1726	0.0570	0.6098	0.3295	0.0408	0.0086
$\mathcal{L}_{cls_2} + \mathcal{L}_{dc} + \mathcal{L}_{apc}$	0.7805	0.6674	0.8730	0.8015	0.8707	0.5621	0.5744	0.0582	0.7850	0.3605	0.3940	0.0102
$\mathcal{L}_{cls_1} + \mathcal{L}_{cls_2}$	0.6149	0.5223	0.7489	0.8005	0.4912	0.5482	0.1207	0.0398	0.2171	0.3680	0.2874	0.0082
$\mathcal{L}_{cls_1} + \mathcal{L}_{cls_2} + \mathcal{L}_{dc}$	0.7045	0.6277	0.8119	0.8012	0.8566	0.5617	0.1807	0.0557	0.4779	0.3705	0.0409	0.0128
$\mathcal{L}_{all}$	0.8094	0.6710	0.9425	0.8019	0.9001	0.5954	0.5874	0.0615	0.8098	0.3748	0.6521	0.0132

**Figure 3: Impact of λ_1 and λ_2 on the Photo, Reddit, Amazon, Elliptic, T-Finance, DGraph datasets. Figure (a) shows AUROC and the Figure (b) shows AUPRC.**

of 0.6710, outperforming GGAD by 5.6%, which can be attributed to its effective modeling of cross-community anomaly propagation. Yet, its AUPRC (0.0615) is slightly lower than GGAD's (0.0610), indicating room for improvement when positive samples are extremely rare. Under extreme class imbalance, as in DGraph, where anomalies constitute less than 0.1% of nodes, PAGE attains an AUPRC of

0.0132, 61% higher than the baseline, demonstrating strong performance under extreme sparsity.

In summary, PAGE captures complex anomaly patterns through progressive attribute-structure perturbation with an anomaly progressive constraint loss. The results confirm its effectiveness across diverse anomaly types and graph characteristics, establishing a new state-of-the-art in semi-supervised GAD.

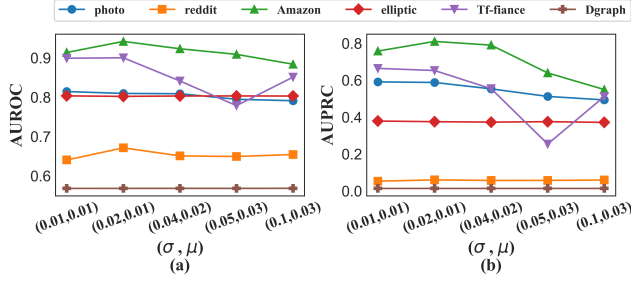


Figure 4: Impact of noise parameters σ and μ of PAGE. Figures (a) and (b) show the overall AUROC and AUPRC trends across noise intensities for each dataset.

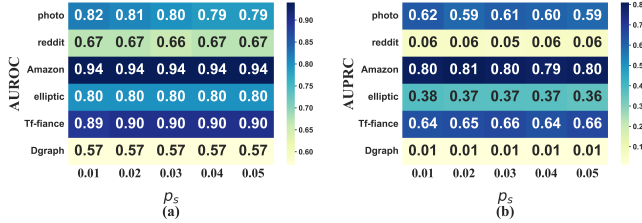


Figure 5: Impact of mask probability p_s . AUROC and AUPRC trends are shown in (a) and (b).

4.3 Ablation Study

To evaluate the individual contributions and synergistic effects of different loss functions, we conducted a comprehensive ablation study. We first examined models trained solely with the binary cross-entropy loss ($\mathcal{L}_{cls_1}$, $\mathcal{L}_{cls_2}$, or their combination), and then progressively incorporated the deviation constraint loss $\mathcal{L}_{dc}$ and the anomaly progressive constraint loss $\mathcal{L}_{apc}$ to enhance representational capacity and discriminative power. As shown in Table 2, the results demonstrate consistent performance gains with the addition of loss components: using only $\mathcal{L}_{cls_1}$ yields 0.1207 AUPRC on Photo, which increases to 0.1807 with $\mathcal{L}_{dc}$ and further surges to 0.5571 when $\mathcal{L}_{apc}$ is introduced. The full model ($\mathcal{L}_{all} = \mathcal{L}_{cls_1} + \mathcal{L}_{cls_2} + \mathcal{L}_{dc} + \mathcal{L}_{apc}$) achieves the best overall performance – for instance, 0.9001 AUROC (+2.7%) and 0.6521 AUPRC (+6.9%) on T-Finance compared with suboptimal variants, while on DGraph, the AUPRC (0.0132) exceeds the best baseline by 61%. These findings verify that the loss components function synergistically through complementary mechanisms: the classification losses establish decision boundaries, the deviation constraint loss preserves topological information, and the progressive constraint refines anomaly discrimination, collectively enabling more effective representation learning and stronger generalization.

4.4 Hyperparameter Sensitivity of PAGE

Impact of λ_1 and λ_2 . For all datasets, we fix $\lambda_3 = 1.5$. The weighting of $\mathcal{L}_{cls_2}$ loss and the deviation constraint loss $\mathcal{L}_{dc}$ are λ_1 and λ_2 , respectively. As shown in Figures 3, the range of λ_1 is $\{1e-6, 1e-5, 1e-4, 1e-3\}$, and the range of λ_2 is $\{0.1, 0.5, 1, 1.5\}$. Increasing λ_2 consistently enhances detection performance, while a moderate $\lambda_1 = 1e-5$ yields optimal and most stable results. Most importantly, the results remain stable across varying values of hyperparameters λ_1 and λ_2 , demonstrating PAGE’s effectiveness.

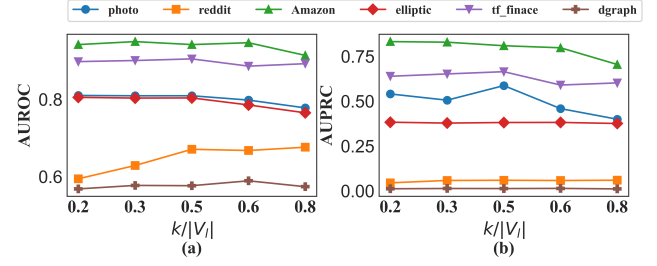


Figure 6: AUROC (a) and AUPRC (b) under different $k/|V_l|$.

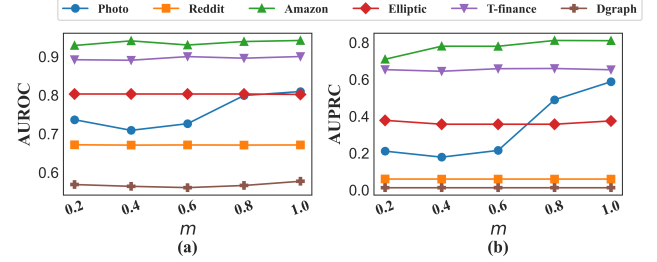


Figure 7: AUROC (a) and AUPRC (b) results with varying m values on six datasets.

Impact of noise parameters σ and μ . The variance σ and mean μ jointly determine the intensity of noise in attribute perturbation. As shown in Figures 4 (a) and (b), models on Photo, T-Finance, Reddit, and DGraph maintain stable performance across different noise levels. However, significant performance degradation occurs on Amazon and T-Finance datasets when noise levels reach (0.05, 0.03), potentially due to these datasets’ sensitivity to noise. Notably, moderate noise levels (0.02, 0.01) yield the most robust and optimal results across all evaluation metrics.

Impact of mask probability p_s . When performing structural perturbations, we delete existing edges and add edges based on the mask probability p_s . This paper constrains the structural perturbation probability p_s within $\{0.01, 0.02, 0.03, 0.04, 0.05\}$. Excessive p_s values would remove informative edges while introducing noisy connections. As evidenced in Figure 5, detection performance remains stable across varying levels of structural perturbations.

Impact of $k/|V_l|$. We systematically select a subset of labeled normal nodes for attribute-structural perturbation. Given $|V_l|$ labeled normal nodes, we perturb k nodes with the selection ratio denoted as $k/|V_l| \in \{0.2, 0.3, 0.5, 0.6, 0.8\}$. As shown in Figure 6, both AUROC and AUPRC remain largely stable across varying selection ratios. On most datasets, performance degrades significantly when $k/|V_l| \geq 0.6$, indicating that excessive perturbation introduces noisy patterns that disrupt the learning of normal node representations. These observations demonstrate that moderate perturbation ratios yield optimal performance. The consistent stability across label ratios empirically validates PAGE’s applicability.

Impact of the m parameter. To evaluate the impact of the parameter m in the anomaly progressive constraint loss, we conducted sensitivity experiments across six datasets. As shown in Figure 7, performance on Reddit and Elliptic remains relatively stable across different m values, suggesting their anomaly patterns are less sensitive to variations in m . In contrast, performance on Photo, Amazon,

Table 3: Performance comparison of different outlier node generation methods on six datasets. The best result for each metric and dataset is highlighted in bold.

Metric	Method	Photo	Reddit	Amazon	Elliptic	T-Finance	DGraph
AUROC	Random	0.5678	0.5227	0.7263	0.6856	0.4613	0.5712
	NLO	0.5307	0.5638	0.8613	0.6787	0.6179	0.5538
	Noise	0.5940	0.5285	0.8508	0.6786	0.8204	0.5779
	GaussianP	0.5925	0.5235	0.2279	0.6715	0.6659	0.5862
	VAE	0.6222	0.6175	0.8984	0.7055	0.6674	0.5801
	GAN	0.6032	0.5378	0.8288	0.6256	0.5487	0.5101
	GGAD	0.6476	0.6354	0.9443	0.7290	0.8228	0.5943
	PAGE	0.8094	0.6710	0.9425	0.8019	0.9001	0.5954
AUPRC	Random	0.1063	0.0394	0.1755	0.1981	0.0402	0.0061
	NLO	0.1092	0.0495	0.4696	0.1750	0.1364	0.0065
	Noise	0.1200	0.0381	0.5384	0.1924	0.1762	0.0076
	GaussianP	0.1194	0.0376	0.0397	0.1682	0.0677	0.0078
	VAE	0.1272	0.0528	0.6111	0.2344	0.0652	0.0063
	GAN	0.1143	0.0433	0.3715	0.1263	0.0461	0.0051
	GGAD	0.1442	0.0610	0.7922	0.2425	0.1825	0.0082
	PAGE	0.5874	0.0615	0.8098	0.3748	0.6521	0.0132

Table 4: Runtimes (in seconds) on the six datasets on CPU.

Method	Photo	Reddit	Amazon	Elliptic	T-Finance	DGraph
DOMINANT	437	125	1592	1119	10721	388
AnomalyDAE	445	161	1656	8296	18560	457
OCGNN	125	162	765	3517	5717	/
AEGIS	417	166	1121	5638	15258	1022
GAAN	307	94	1678	1866	12120	/
TAM	165	432	4516	13200	17360	/
GGAD	106	368	658	5146	9345	488
PAGE (Ours)	408	608	951	2180	1668	1176

and DGraph is more sensitive to changes in m , generally improving as m increases to 0.8 or 1.0, especially for larger datasets. For T-Finance, results remain stable across settings, with minor improvements at $m = 0.6$ and $m = 1.0$. Taken together, $m = 0.8$ or 1.0 consistently delivers superior or more stable detection performance across most datasets, demonstrating its strong generalizability.

Comparison of Outlier Node Generation Methods. To further validate the outlier node generation module in GGAD, we compare several alternative strategies, as shown in Table 3. (i) **Random**: randomly samples normal nodes as outliers to train a one-class classifier, but these outliers are difficult to distinguish from normal nodes, resulting in poor performance. (ii) **NLO (Non-learnable Outliers)**: removes learnable parameters from GGAD’s module; it performs reasonably on some datasets (e.g., Amazon, T-Finance, and Elliptic) but overall is inferior, indicating that learnable representations better capture anomaly characteristics. (iii) **GaussianP (gaussian perturbation)**: generates outliers via random noise or gaussian perturbations of normal node embeddings; while separable from normal nodes, the lack of structural information causes a distribution mismatch with real anomalies. (iv) **VAE and GAN**: generate anomalies using variational autoencoders and generative adversarial networks with anomaly prior constraints. Based on these observations, our method innovates further by proposing a *progressive anomaly generation strategy*, which gradually increases the severity of anomalies. This approach enables more fine-grained anomaly supervision.

4.5 Computational Efficiency Analysis

Time Complexity Analysis. Given a graph G with n nodes and feature dimension M , we first propagate it through a graph neural network (GNN) to obtain updated node representations $\mathbf{H} \in \mathbb{R}^{n \times d_1}$, then inject feature noise into a subset of k target nodes, perform structural perturbations on their local adjacency structures, and compute three objective functions: (1) deviation constraint loss for the k perturbed nodes; (2) progressive anomaly constraint loss for the k perturbed nodes; and (3) classification loss over all n nodes.

We analyze the time complexity of PAGE as follows. Feature noise injection requires $\mathcal{O}(kd_1)$ operations, structural perturbation involves traversing the neighbors of k nodes and results in $\mathcal{O}(k \cdot e_{\max})$ complexity, where $e_{\max}$ denotes the maximum node degree, and loss computation consists of three parts: the deviation constraint loss for k nodes with complexity $\mathcal{O}(kd_1)$, the anomaly progressive constraint loss with complexity $\mathcal{O}(kd_1)$, and the global classification loss with complexity $\mathcal{O}(nC)$, where C is the number of classes. Consequently, the overall time complexity is $\mathcal{O}(kd_1 + ke_{\max} + nC)$, which scales linearly with k and n .

Runtime Results. Computational costs are summarized in Table 4. We evaluate all methods on six datasets. PAGE maintains competitive detection performance while operating within acceptable time budgets. For large-scale datasets (e.g., T-Finance and Elliptic), PAGE requires only 1668 and 2180 seconds, demonstrating remarkable scalability. On Photo, Reddit, and Amazon, PAGE incurs moderately higher runtime than GGAD with marginal increments, yet achieves better detection accuracy. Notably, PAGE successfully trains on structurally complex graph DGraph, whereas several baselines fail to execute. Collectively, PAGE demonstrates both computational efficiency and scenario adaptability across diverse graph.

5 Conclusion

In this paper, we propose PAGE, a novel framework for semi-supervised GAD under the challenging “normal-only labels” setting. PAGE addresses the critical limitation of existing methods: their failure to model attribute-structure synergy during anomaly generation through progressive attribute-structure perturbation. PAGE first injects attribute noise into normal nodes to generate primary anomalies, then applies structural perturbations (e.g., edge rewiring) to create hybrid anomalies that simulate real-world anomaly evolution. Crucially, we design an anomaly progressive constraint loss that enforces a strict abnormality hierarchy while quantifying anomaly severity through triplet distance constraints in the embedding space, thus enabling fine-grained anomaly characterization. Extensive experiments confirm that PAGE outperforms state-of-the-art baselines. In future work, we will extend PAGE to dynamic graph adaptation and explore the theoretical foundations for attribute-structure synergy in anomaly generation.

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References

- [1] Leman Akoglu, Hanghang Tong, and Danai Koutra. 2015. Graph based anomaly detection and description: a survey. *Data mining and knowledge discovery* 29, 3 (2015), 626–688.
- [2] Bo Chen, Jing Zhang, Xiaokang Zhang, Yuxiao Dong, Jian Song, Peng Zhang, Kaibo Xu, Evgeny Kharlamov, and Jie Tang. 2022. Gccad: Graph contrastive coding for anomaly detection. *IEEE Transactions on Knowledge and Data Engineering* 35, 8 (2022), 8037–8051.
- [3] Nan Chen, Zemin Liu, Bryan Hooi, Bingsheng He, Rizal Fathony, Jun Hu, and Jia Chen. 2024. Consistency Training with Learnable Data Augmentation for Graph Anomaly Detection with Limited Supervision. In *The Twelfth International Conference on Learning Representations*.
- [4] Zhenxing Chen, Bo Liu, Meiqing Wang, Peng Dai, Jun Lv, and Liefeng Bo. 2020. Generative adversarial attributed network anomaly detection. In *Proceedings of the 29th ACM International Conference on Information & Knowledge Management*. 1989–1992.
- [5] Junwei Cheng, Ke Liang, Weixiong Liu, Pengxing Feng, and Chaobo He. 2026. Multi-Scale Signal Modulation for Variational Graph Autoencoders. *Artificial Intelligence* (2026), 104488.
- [6] Jiafeng Cheng, Qianqian Wang, Zhiqiang Tao, Deyan Xie, and Quanxue Gao. 2021. Multi-view attribute graph convolution networks for clustering. In *Proceedings of the twenty-ninth international conference on international joint conferences on artificial intelligence*. 2973–2979.
- [7] Kaize Ding, Jundong Li, Nitin Agarwal, and Huan Liu. 2021. Inductive anomaly detection on attributed networks. In *Proceedings of the twenty-ninth international conference on international joint conferences on artificial intelligence*. 1288–1294.
- [8] Kaize Ding, Jundong Li, Rohit Bhanushali, and Huan Liu. 2019. Deep anomaly detection on attributed networks. In *Proceedings of the 2019 SIAM international conference on data mining*. 594–602.
- [9] Yingdong Dou, Zhiwei Liu, Li Sun, Yutong Deng, Hao Peng, and Philip S Yu. 2020. Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In *Proceedings of the 29th ACM international conference on information & knowledge management*. 315–324.
- [10] Jingcan Duan, Siwei Wang, Pei Zhang, En Zhu, Jingtao Hu, Hu Jin, Yue Liu, and Zhibin Dong. 2023. Graph anomaly detection via multi-scale contrastive learning networks with augmented view. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 7459–7467.
- [11] Ocheme Anthony Ekle and William Eberle. 2024. Anomaly detection in dynamic graphs: A comprehensive survey. *ACM Transactions on Knowledge Discovery from Data* 18, 8 (2024), 1–44.
- [12] Haoyi Fan, Fengbin Zhang, and Zuoyong Li. 2020. Anomalydae: Dual autoencoder for anomaly detection on attributed networks. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 5685–5689.
- [13] Chenghao Fang, Jianqing Liang, Jiye Liang, Hangkun Wang, Kaixuan Yao, and Feilong Cao. 2025. Multi-modal point cloud completion with interleaved attention enhanced Transformer. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. 963–971.
- [14] Yuan Gao, Xiang Wang, Xiangnan He, Zhengguang Liu, Huamin Feng, and Yongdong Zhang. 2023. Addressing heterophily in graph anomaly detection: A perspective of graph spectrum. In *Proceedings of the ACM web conference 2023*. 1528–1538.
- [15] Yuan Gao, Xiang Wang, Xiangnan He, Zhengguang Liu, Huamin Feng, and Yongdong Zhang. 2023. Alleviating structural distribution shift in graph anomaly detection. In *Proceedings of the sixteenth ACM international conference on web search and data mining*. 357–365.
- [16] T. Guo, D. Wang, J. Liang, K. Zhang, and J. Zeng. 2024. SpeAr: A Spectral Approach for Zero-Shot Node Classification. In *Advances in Neural Information Processing Systems*. 69000–69021.
- [17] Junwei He, Qianqian Xu, Yangbangyan Jiang, Zitai Wang, and Qingming Huang. 2024. Ada-gad: Anomaly-denoised autoencoders for graph anomaly detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 8481–8489.
- [18] Mengda Huang, Yang Liu, Xiang Ao, Kuan Li, Jianfeng Chi, Jinghua Feng, Hao Yang, and Qing He. 2022. Auc-oriented graph neural network for fraud detection. In *Proceedings of the ACM web conference 2022*. 1311–1321.
- [19] Xuanwen Huang, Yang Yang, Yang Wang, Chunping Wang, Zhisheng Zhang, Jiarong Xu, Lei Chen, and Michalis Vazirgiannis. 2022. Dgraph: A large-scale financial dataset for graph anomaly detection. *Advances in Neural Information Processing Systems* 35 (2022), 22765–22777.
- [20] Pengfei Jiao, Tianpeng Li, Yingjie Xie, Yinghui Wang, Wenjun Wang, Dongxiao He, and Huaming Wu. 2021. Generative evolutionary anomaly detection in dynamic networks. *IEEE Transactions on Knowledge and Data Engineering* 35, 12 (2021), 12234–12248.
- [21] Srikanth Kumar, Xikun Zhang, and Jure Leskovec. 2019. Predicting dynamic embedding trajectory in temporal interaction networks. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*. 1269–1278.
- [22] Yonghyeon Lee, Hyeokjun Kwon, and Frank Park. 2021. Neighborhood reconstructing autoencoders. *Advances in Neural Information Processing Systems* 34 (2021), 536–546.
- [23] Ke Liang, Lingyuan Meng, Hao Li, Jun Wang, Long Lan, Miaomiao Li, Xinwang Liu, and Huamin Wang. 2025. From Concrete to Abstract: Multi-View Clustering on Relational Knowledge. *IEEE Trans. Pattern Anal. Mach. Intell.* 47, 10 (2025), 9043–9060.
- [24] Fanzhen Liu, Xiaoxiao Ma, Jia Wu, Jian Yang, Shan Xue, Amin Beheshti, Chuan Zhou, Hao Peng, Quan Z Sheng, and Charu C Aggarwal. 2022. Dagad: Data augmentation for graph anomaly detection. In *2022 IEEE international conference on data mining (ICDM)*. 259–268.
- [25] Yang Liu, Xiang Ao, Zidi Qin, Jianfeng Chi, Jinghua Feng, Hao Yang, and Qing He. 2021. Pick and choose: a GNN-based imbalanced learning approach for fraud detection. In *Proceedings of the web conference 2021*. 3168–3177.
- [26] Xiaoxiao Ma, Ruikun Li, Fanzhen Liu, Kaize Ding, Jian Yang, and Jia Wu. 2024. Graph anomaly detection with few labels: A data-centric approach. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2153–2164.
- [27] Julian McAuley, Christopher Targett, Qinfeng Shi, and Anton Van Den Hengel. 2015. Image-based recommendations on styles and substitutes. In *Proceedings of the 38th international ACM SIGIR conference on research and development in information retrieval*. 43–52.
- [28] Hezhe Qiao and Guansong Pang. 2023. Truncated affinity maximization: One-class homophily modeling for graph anomaly detection. *Advances in Neural Information Processing Systems* 36 (2023), 49490–49512.
- [29] Hezhe Qiao, Hanghang Tong, Bo An, Irwin King, Charu Aggarwal, and Guansong Pang. 2024. Deep graph anomaly detection: A survey and new perspectives. *arXiv preprint arXiv:2409.09957* (2024).
- [30] Hezhe Qiao, Qingsong Wen, Xiaoli Li, Ee-Peng Lim, and Guansong Pang. 2024. Generative semi-supervised graph anomaly detection. *Advances in neural information processing systems* 37, 4660–4688.
- [31] Chenyang Qiu, Zhaoci Huang, Wenzhe Xu, and Huijia Li. 2022. VGAER: graph neural network reconstruction based community detection. *arXiv preprint arXiv:2201.04066* (2022).
- [32] Amit Roy, Juan Shu, Jia Li, Carl Yang, Olivier Elshocht, Jeroen Smets, and Pan Li. 2024. Gad-nr: Graph anomaly detection via neighborhood reconstruction. In *Proceedings of the 17th ACM international conference on web search and data mining*. 576–585.
- [33] Fengzhao Shi, Yanan Cao, Yanmin Shang, Yuchen Zhou, Chuan Zhou, and Jia Wu. 2022. H2-fdetector: A gnn-based fraud detector with homophilic and heterophilic connections. In *Proceedings of the ACM web conference 2022*. 1486–1494.
- [34] Chuancheng Song, Xixun Lin, Hanyang Shen, Yanmin Shang, and Yanan Cao. 2025. Uniform: Towards unified framework for anomaly detection on graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 12559–12567.
- [35] Jianheng Tang, Jiajin Li, Ziqi Gao, and Jia Li. 2022. Rethinking graph neural networks for anomaly detection. In *International conference on machine learning*. 21076–21089.
- [36] Li Wang and Ren-cang Li. 2019. Learning low-dimensional latent graph structures: A density estimation approach. *IEEE Transactions on Neural Networks and Learning Systems* 31, 4 (2019), 1098–1112.
- [37] Qizhou Wang, Guansong Pang, Mahsa Salehi, Xiaokun Xia, and Christopher Leckie. 2025. Open-Set Graph Anomaly Detection via Normal Structure Regularisation. In *International conference on learning representations*.
- [38] Xuhong Wang, Baihong Jin, Ying Du, Ping Cui, Yingshui Tan, and Yupu Yang. 2021. One-class graph neural networks for anomaly detection in attributed networks. *Neural computing and applications* 33, 18 (2021), 12073–12085.
- [39] Mark Weber, Giacomo Domeniconi, Jie Chen, Daniel Karl I Weidele, Claudio Bellei, Tom Robinson, and Charles E Leiserson. 2019. Anti-money laundering in bitcoin: Experimenting with graph convolutional networks for financial forensics. *arXiv preprint arXiv:1908.02591* (2019).
- [40] Qin Xie, Peng Zhang, Boseon Yu, and Jaesik Choi. 2021. Semisupervised training of deep generative models for high-dimensional anomaly detection. *IEEE Transactions on Neural Networks and Learning Systems* 33, 6 (2021), 2444–2453.
- [41] Xihong Yang, Yue Liu, Sihang Zhou, Siwei Wang, Wenxuan Tu, Qun Zheng, Xinwang Liu, Liming Fang, and En Zhu. 2023. Cluster-guided contrastive graph clustering network. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 37. 10834–10842.
- [42] Yu Zheng, Ming Jin, Yixin Liu, Lianhua Chi, Khoa T Phan, and Yi-Ping Phoebe Chen. 2021. Generative and contrastive self-supervised learning for graph anomaly detection. *IEEE Transactions on Knowledge and Data Engineering* 35, 12 (2021), 12220–12233.
- [43] Qinghai Zhou, Yuzhong Chen, Zhe Xu, Yuhang Wu, Menghai Pan, Mahashweta Das, Hao Yang, and Hanghang Tong. 2024. Graph anomaly detection with adaptive node mixup. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 3494–3504.